TUSHAR SINGH

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SUMMARY

AI/ML and **Backend Engineer** with hands-on experience shipping Retrieval-Augmented Generation(RAG) systems, computer-vision pipelines, and full-stack applications across 7+ national-level hackathons. Skilled in **Python**, **Java**, and **FastAPI**, with working knowledge of **LangChain**, **YOLOv8/YOLOv11**, and modern **LLMs** (Llama 3.3, Gemini); currently deepening expertise in Deep Learning en route to Generative AI and Agentic AI. **B.Tech CSE undergraduate (AKTU)** seeking a Backend Development / AI-ML Internship.

TECHNICAL SKILLS

Languages: Python, Java, C

Backend & Web: FastAPI, Flask, Django, Streamlit, WebSocket

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB, FAISS

AI / ML: Machine Learning, Deep Learning, Generative AI, RAG, Embeddings, Agentic AI, LangChain, Prompt Engineering, LLMs (Llama 3.3, Gemini), YOLOv8/YOLOv11, OpenCV, Google MediaPipe, HuggingFace

Tools & Platforms: Docker, Linux, Git/GitHub, Supabase, Groq API, Google Gemini API, ZegoCloud WebRTC

PROJECTS

JurixAI — RAG-Based AI Legal Assistant

LangChain, ChromaDB, Groq/Llama-3.3-70B, HuggingFace

- Engineered a **Retrieval-Augmented Generation legal chatbot** covering Indian statutes (IPC, POSH, RTI, Consumer Protection) with Llama-3.3-70B via Groq Cloud; semantic retrieval pipeline (HuggingFace all-MiniLM-L6-v2 + ChromaDB) **returns top-8 relevant contexts** with source citations.
- Delivered bilingual (English/Hinglish) responses via a custom empathy-first prompting strategy; added **OCR support** (pdfplumber, pytesseract) for scanned document uploads.

Antriksha Search Engine (ASE) — Cross-Modal Satellite Image Retrieval

ISRO Bharatiya Antariksh Hackathon 2026 · RemoteCLIP, FAISS, PyTorch

- Designed a **geospatial AI platform bridging SAR and optical satellite imagery using RemoteCLIP (ViT-B/32) for cross-modal + text-to-image vector search**; indexed 32,000+ embeddings in FAISS at ~0.04s retrieval latency.
- Led a 4-member team (Syntax Squad) as Lead ML Architect, owning system orchestration and vector database architecture.

RoadGuard AI — AI-Powered Pothole Detection & Road Management | *Smart India Hackathon (SIH) 2026 · Next.js, FastAPI, YOLO, Docker*

- Architected a full-stack platform (Next.js/React frontend, FastAPI backend) that detects and **geotags potholes from live camera/drone feeds using a custom-trained YOLO model**, then automates severity classification → cost estimation → PDF report → email dispatch to authorities.
- Owned full-stack development and AI integration deployed production infrastructure with Docker, and Caddy (auto-HTTPS) on a Microsoft Azure VM.

HazardLens — Real-Time Edge AI Hazard Detection Platform | *Build With Bharat (Microsoft) · YOLOv8, OpenCV, Edge-TTS, Streamlit*

- Developed an **edge-optimized surveillance system using dual fine-tuned YOLOv8 models to detect fire, smoke, and crowd overcrowding from live CCTV/IP feeds**, with an offline Hindi neural voice alert system (Edge-TTS) and a confidence-tuned (0.70) false-positive mitigation layer.

SignBridge AI — Real-Time Sign Language Translation Platform (in progress)

| FastAPI, WebSocket, MediaPipe, Gemini, ZegoCloud

- Implementing a **real-time communication tool for the hearing/speech impaired** using Google MediaPipe (21-point hand tracking, 64K-image dataset) and Gemini Flash for sentence generation, with ZegoCloud WebRTC video calling and a built-in sign-language teaching module.

EDUCATION

Meerut Institute of Engineering & Technology (MIET), AKTU — **B.Tech, CSE, 3rd Year** with current CGPA of **7.63** ***(2024 – 2028)***

Rishabh Academy — Intermediate **(PCM), 77%** ***(2023)***

Karan Public School — High School, **77.8%** ***(2020)***

ACHIEVEMENTS

- Reached the final round at **6 national-level** hackathons: Smart India Hackathon (SIH) 2025 & 2026, Build With Bharat 1.0 (Microsoft), HackSynergy, Bharat Expo 2026, LITHOS, and DevGathering.
- Architected and deployed a cross-modal satellite search engine (ASE) at the ISRO Bharatiya Antariksh Hackathon 2026 as Lead ML Architect, clearing Round 1.

CERTIFICATIONS

- Machine Learning Training Certification — Meerut Institute of Engineering and Technology.
- Multiple hackathon participation certificates – (SIH, Build With Bharat, HackSynergy, Bharat Expo, LITHOS, DevGathering, ISRO BAH2026)